an accuracy of better than one part in a billion. This is an absolutely astonishing agreement between abstract theoretical calculation and the real world. Through quantum electrodynamics, physicists have been able to solidify the role of photons as the “smallest possible bundles of light” and to reveal their interactions with electrically charged particles such as electrons, in a mathematically complete, predictive, and convincing framework.

The success of quantum electrodynamics inspired other physicists in the 1960s and 1970s to try an analogous approach for developing a quantum-mechanical understanding of the weak, the strong, and the gravitational forces. For the weak and the strong forces, this proved to be an immensely fruitful line of attack. In analogy with quantum electrodynamics, physicists were able to construct quantum field theories for the strong and the weak forces, called quantum chromodynamics and quantum electroweak theory. “Quantum chromodynamics” is a more colorful name than the more logical “quantum strong dynamics,” but it is just a name without any deeper meaning; on the other hand, the name “electroweak” does summarize an important milestone in our understanding of the forces of nature.

Through their Nobel Prize-winning work, Sheldon Glashow, Abdus Salam, and Steven Weinberg showed that the weak and electromagnetic forces are naturally united by their quantum field-theoretic description even though their manifestations seem to be utterly distinct in the world around us. After all, weak force fields diminish to almost vanishing strength on all but subatomic distance scales, whereas electromagnetic fields—visible light, radio and TV signals, X-rays—have an indisputable macroscopic presence. Nevertheless, Glashow, Salam, and Weinberg showed, in essence, that at high enough energy and temperature—such as occurred a mere fraction of a second after the big bang—electromagnetic and weak force fields dissolve into one another, take on indistinguishable characteristics, and are more accurately called electroweak fields. When the temperature drops, as it has done steadily since the big bang, the electromagnetic and weak forces crystallize out in a different manner from their common high-temperature form—through a process known as symmetry breaking that we will describe later—and